

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/271464215>

Continuous control of blood glucose in T1DM using fuzzy logic controller in insulin pump: A simulation study

Conference Paper · December 2011

DOI: 10.1109/ICCIAutom.2011.6356642

CITATIONS

6

2 authors:



Ali Maleki

Semnan University

52 PUBLICATIONS 102 CITATIONS

[SEE PROFILE](#)

READS

115



Arezou Geramipour

Florida International University

5 PUBLICATIONS 15 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Effects of stress on moral judgment [View project](#)



FES IN HAND FOR REACHING [View project](#)

Continuous Control of Blood Glucose in T1DM Using Fuzzy Logic Controller in Insulin Pump: A Simulation Study

Ali Maleki and Arezou Gheramipour

Abstract—this paper presents a closed-loop control system for regulating blood glucose level in type-1 diabetic patients. The goals of designing controller are 1. avoidance of hypoglycemia (blood glucose concentrations should be above 3mmol/l) and 2. reduction in the blood glucose level (average blood glucose should be less than 7mmol/l) in diabetic patients. An intensive therapy for these patients is described by a continuous subcutaneous insulin infusion pump which mimics the natural insulin secretion in healthy person. The control algorithm uses experience of physicians to treat this disease by using Mamdani-type fuzzy logic controller (FLC). Robustness of this controller is tested by a group of diabetic patients models. Then to assess the performance and capabilities of fuzzy controller, the results of FLC were compared to PID controller for each diabetic patient. The results showed that FLC has a more stable response in presence of uncertainty in system parameters which can vary from patient to patient.

I. INTRODUCTION

Diabetes mellitus is a metabolic disease in which the body does not produce or properly use insulin. The pancreatic hormones insulin and glucagon which are secreted from β -cells and α -cells respectively are responsible for keeping the blood glucose concentration level in a normal range.

There are two types of diabetes: type I and type II. Type-I diabetes mellitus (T1DM) is a disease characterized by absence or death of the β -cells in the pancreatic islets of Langerhans. High average glucose concentration (above 7 mmol/L) is called Hyperglycemia. Since the β -cells are responsible to produce insulin, external insulin injections in T1DM are required to regulate the blood glucose concentration. Hypoglycemia is a complication that occurs for T1DM when the blood glucose level is too low (less than 2 mmol/L [1]) which is the result of too high injected insulin and is very dangerous if it is not rapidly treated with glucose intake.

In a typical diabetic treatment, patient measures own blood glucose level 3 or 4 times in a day and attempts to maintain it in a normal range by injecting appropriate rate of insulin. The size of the injections is based on the blood

glucose measurements, knowledge about food intake and to a large part on the personal experience of the individual diabetic [2]. An insulin pump is a small device which is attached to patient. It consists of a controller (determines appropriate dose of insulin), a reservoir (for storing insulin) and a cannula (for infusion insulin into body). Sensor measures the blood glucose and transmits this information to the controller which is implanted in pump. Then the controller determines appropriate dose of insulin.

There are several control algorithms for regulating blood glucose level. One of the traditional controllers is PID controller. Steil *et al.* in 2004 used a couple of PID controllers. Their results were that this controller “closely parallels the β -cell biphasic insulin response” [3]. One of the disadvantages of PID controller is that it is not considered the changes in patients parameters which can vary from patient to patient. Another control technique is model reference adaptive control. Wahab *et al.* in 2006 used Bolie model for doing model reference adaptive control [4]. In model predictive control (MPC), the model of patient is required. Hovorka *et al.* in 2004 used the Hovorka model and the MPC approach for regulating blood glucose level [5].

Due to complexity of insulin-glucose system, the aim of this article is to design a fuzzy logic controller (FLC). One of the attractive features of FLCs is their less sensitivity to system parameters that vary from patient to patient. Because an intensive therapy can reduce the morbidity of diabetes illness in long term [6], this type of therapy is suggested for type-I diabetes mellitus. Intensive insulin therapy can be categorized to continuous subcutaneous insulin infusion (CSII) pump or multiple daily injections regimen (MDIR). Kardar *et al.* in 2008 used MDIR method for control diabetes using fuzzy logic controller [7]. In this study, we considered CSII method using insulin pump. The control objective is shown in TABLE I.

In section 2 the physiological model of glucose-insulin regulatory system in T1DM is introduced. Section 3 includes the design of fuzzy controller and PID controller. In section 4 simulation results of fuzzy controller are presented and compared to corresponding results of PID controller. Finally, the paper is finished with the interpretation and discussion of the results.

TABLE I
CONTROL OBJECTIVE FOR GLUCOSE-INSULIN SYSTEM [8]

Glucose	Normal	Target
---------	--------	--------

A. Maleki is in Semnan University, Electrical and Computer Engineering Faculty, Biomedical Engineering Department. Email: ali_maleki@aut.ac.ir.

A. Gheramipour is in Amirkabir University of Technology, Biomedical Engineering Faculty, Biological Systems Control Laboratory. Email: gheramipour@aut.ac.ir.

Before eating	<110	90<<130
Two hours after eating	<110	<180
bedtime	<120	110<<150

II. GLUCOSE-INSULIN SYSTEM MODEL

Although complicated models are accurate to describe glucose-insulin metabolism, but in many cases a simple model would be sufficient to make a good analysis. A simple method with few parameters was introduced by Richard N. Bergman and is called Bergman's minimal model. The model equations are [9]:

$$\begin{aligned} \frac{dG(t)}{dt} &= -p_1[G(t) - G_b] - X(t)G(t) + D(t) \\ \frac{dX(t)}{dt} &= -p_2X(t) + p_3[I(t) - I_b] \\ \frac{dI(t)}{dt} &= -n[I(t) - I_b] + \gamma[G(t) - h]^+ t + U(t) \end{aligned} \quad (1)$$

Where $G(t)$ is the plasma glucose concentration over basal (mg/dl), G_b is the basal plasma glucose (mg/dl), $X(t)$ is the effect of active insulin (1/min), $I(t)$ is the insulin concentration above basal (mU/L), I_b is the basal insulin level (mU/L), $D(t)$ is the rate of mg glucose pr. dl entering the blood. $U(t)$ is the external insulin input (mU/L/min).

p_1 , p_2 , p_3 , n , h and γ are parameters of Bergman minimal model. γ is rate of pancreatic release of insulin after glucose bolus [(mU/L) (min)⁻¹ (mg/dl)⁻¹]. n is decay rate of blood insulin (1/min), and h is the threshold value of blood glucose above which pancreatic β -cells release insulin (mg/dl). The term $\gamma[G(t) - h]^+$ demonstrates insulin secretion in the body which does not exist in T1DM. The clinical data express that the value of p_1 parameter in diabetic patients reduced extremely and can be approximated to zero.

A simple description of $D(t)$ was suggested by Fisher and looks like equation (2) [10]:

$$D(t) = B \exp(-\alpha t), B > 0 \quad (2)$$

III. DESIGN OF FUZZY AND PID CONTROLLERS

The controller is designed with Mamdani-type fuzzy structure, using two input variables and one output variable. The input variables are error and rate of it, and the output variable is rate of insulin infusion. By the definition of the input and output fuzzy sets, a total 49 IF-THEN rules were defined. The shapes of inputs and output membership functions are presented in Fig 1 and the surface of fuzzy inference system is presented in Fig 2. The 49 rules of fuzzy rule base are summarized in Table II.

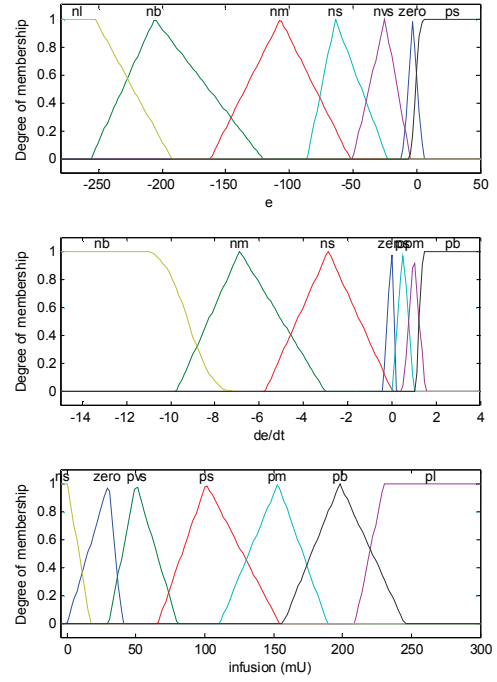


Fig. 1. Membership functions of FLC. The first panel is the membership functions of input “e(t)”, the second panel is the membership functions of input “de(t)/dt” and the third panel shows the membership functions of output “infusion”.

TABLE II
THE RULES OF FUZZY CONTROLLER

E\E	NB	NM	NS	ZERO	PS	PM	PB
NL	PL	PL	PL	PB	PB	PM	PM
NB	PB	PB	PB	PM	PM	PM	PS
NM	PB	PB	PB	PM	PS	PS	PVS
NS	PB	PB	PB	PVS	PVS	PVS	ZERO
NVS	PM	PM	PVS	ZERO	ZERO	ZERO	NS
ZERO	PS	PVS	PVS	NS	NS	NS	NS
PS	NS	NS	NS	NS	NS	NS	NS

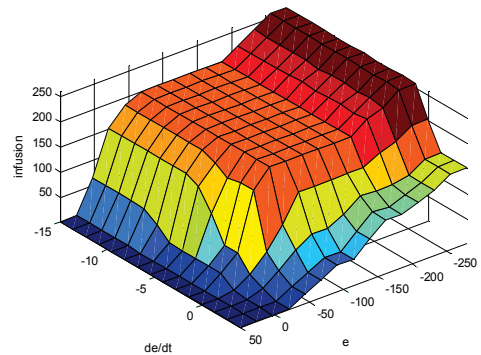


Fig. 2. The surface of fuzzy inference system.

Then, to assess the performance of the FLC in comparison with PID, a PID controller for patient 6 is designed. The parameters of this controller is tuned based on Zigler-Nicols technique and are optimized with try and error approach. The parameters are: $K_p = 7$, $K_i = 0.0032$, $K_d = 250$.

IV. RESULTS

In this section, glucose level control of diabetic patients using fuzzy logic controller was simulated in Matlab/Simulink. Three exogenous glucose meals were considered per day: breakfast (7:00), lunch (13:00) and dinner (21:00). Also, a second order transfer function was used as model of pump. In this paper, we use the oral glucose tolerance test (OGTT) as the exogenous glucose input. It was assumed that all patients began to eat meals after a long time (8 hours) after using insulin pump. Then to assess the performance and capabilities of fuzzy controller, the results of FLC were compared to PID for each diabetic patient. The values of parameters of patients 1, 2 and 3 are in [11] and the parameters of other patients are in [9], [12].

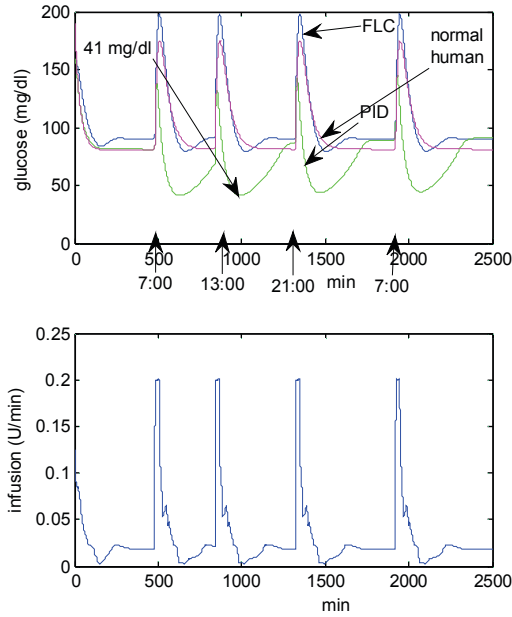


Fig. 3. Simulation results for patient 1 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

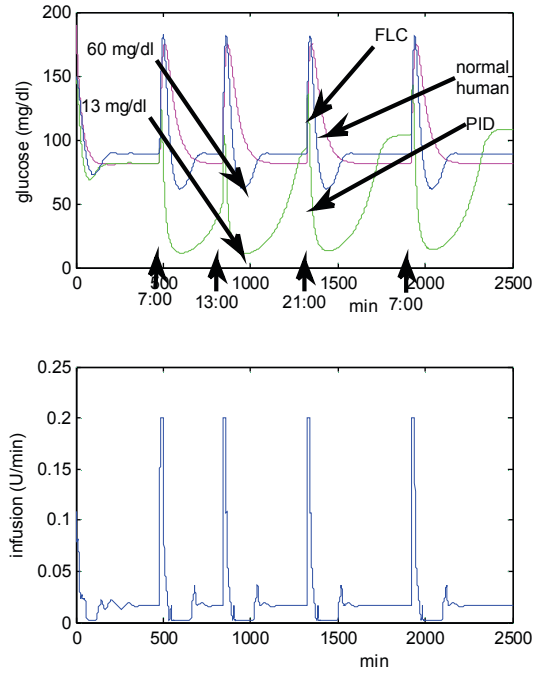


Fig. 4. Simulation results for patient 2 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

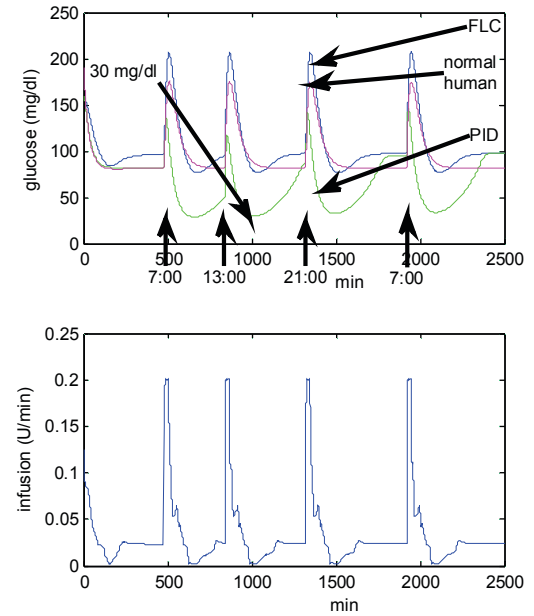


Fig. 5. Simulation results for patient 3 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

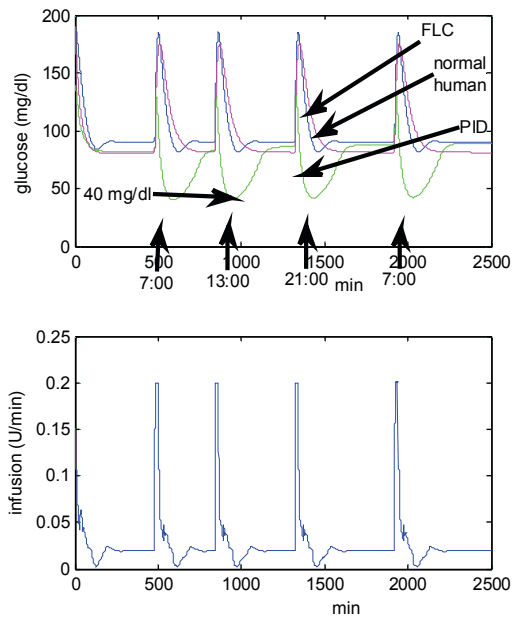


Fig. 6. Simulation results for patient 4 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

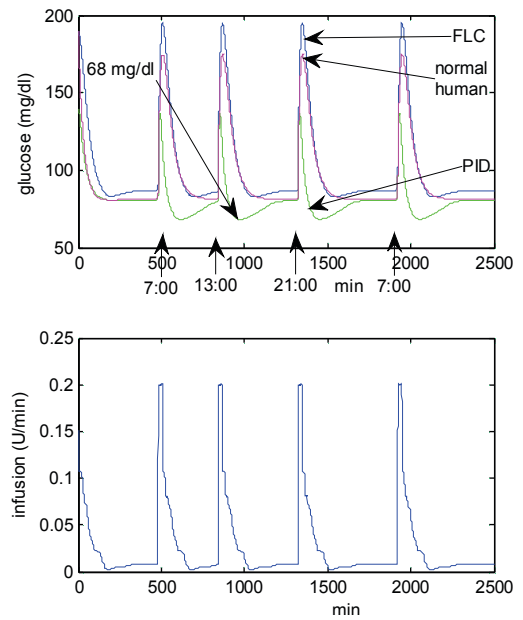


Fig. 8. Simulation results for patient 6 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

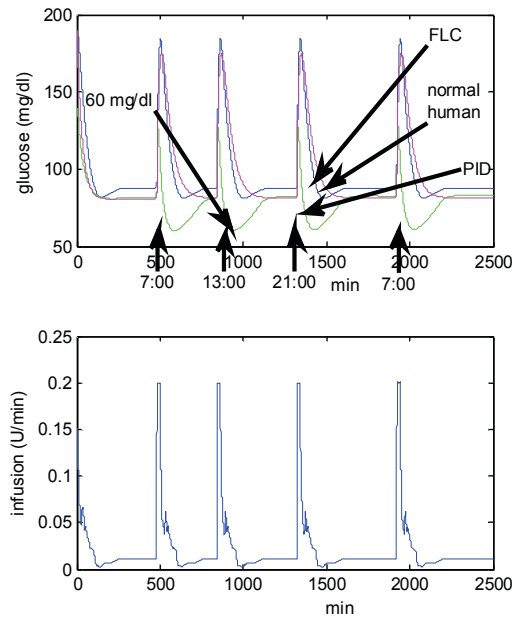


Fig. 7. Simulation results for patient 5 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

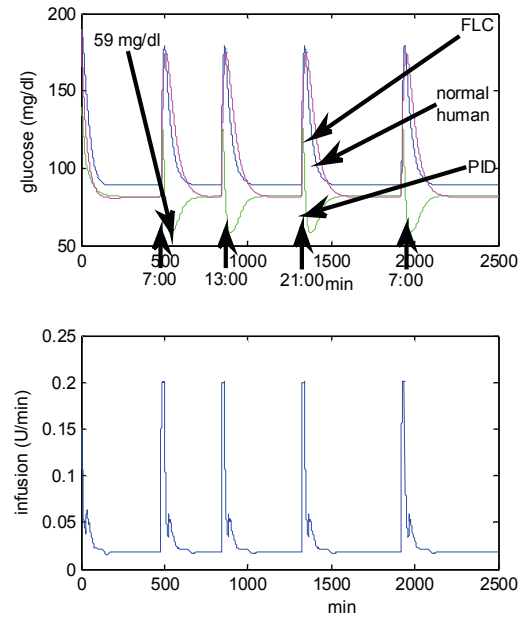


Fig. 9. Simulation results for patient 7 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

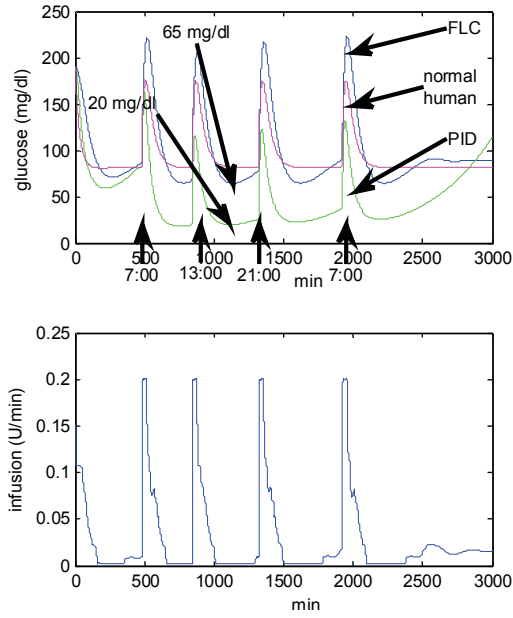


Fig. 10. Simulation results for patient 8 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

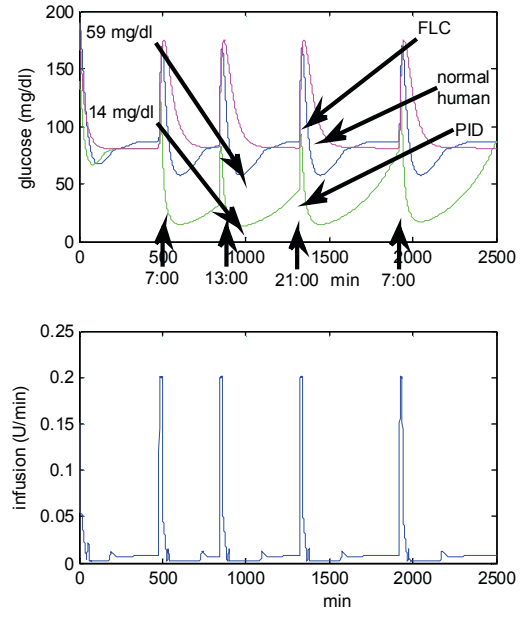


Fig. 12. Simulation results for patient 10 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

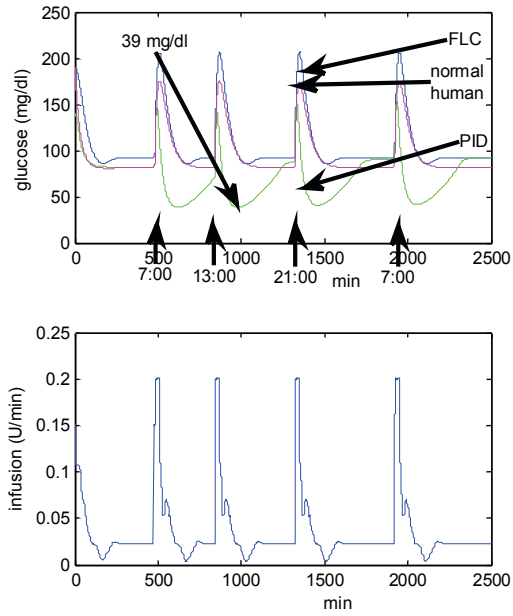


Fig. 11. Simulation results for patient 9 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

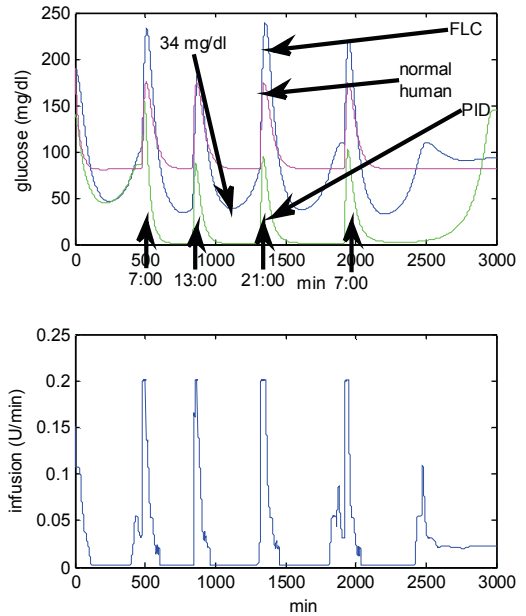


Fig. 13. Simulation results for patient 11 using FLC. The first panel shows blood glucose concentration and the second panel shows infusion of insulin.

TABLE III

TWO QUANTITATIVE CRITERIA OF CONTROLLER PERFORMANCE ASSESSED FOR 11 PATIENTS WITH DIFFERENT INSULIN SENSITIVITY.

Patient description		Minimum(mg/dl)		Two hours after eating(mg/dl)	
Patient #	Insulin sensitivity($\times 10^4$)	PID	FLC	PID	FLC
Patient 1	4.77	41	80	43	97
Patient 2	5.2	13	60	13	62
Patient 3	7	30	77	34	100
Patient 4	5.6	40	80	40	85
Patient 5	5.1	60	80	60	88
Patient 6	3.1	68	83	69	104
Patient 7	2.4	59	90	71	90
Patient 8	2.1	20	65	35	128
Patient 9	2	39	85	40	105
Patient 10	6.5	14	59	15	60
Patient 11	9.6	0	34	9	93

V. DISCUSSION

In this study a fuzzy logic controller for maintaining the blood glucose level of type-I diabetic patients in a normal range, has been designed. Robustness of this controller was tested by a group of diabetic patients with different parameters. As figures Fig. 3 – Fig. 13 illustrate, FLC could reduce blood glucose level of each diabetic patient in appropriate settling time.

One of the important subjects in diabetic patients when they use insulin pump is that the controller which is implemented in pump should be capable to avoiding hypoglycemic states that are the result of a rapid fall of the glucose curve through the normal range [13]. The results of FLC were compared to PID controller. Figures show that FLC could reduce blood glucose of all patients according to American Diabetes Association (ADA) standards and its results are better than PID controller (in terms of hypoglycemia and hyperglycemia prevention) in general. Simulation results showed that peak of blood glucose in all patients when they use PID controller, is lower than the state in which they use fuzzy controller. Indeed the target of PID controller is peak reduction of blood glucose and for this reason it injects a large amount of insulin which leads to rapid fall of blood glucose and hypoglycemic states. But as it is seen in Table I, the goal of diabetic treatment isn't peak reduction of blood glucose.

Insulin sensitivity is a factor that explains how much one unit of insulin lowers blood glucose. In simulation we see 1 state (patient 11) in which the usage of FLC and PID controller lead to hypoglycemia state and it is because insulin sensitivity in patient 11 is larger than other patients and so he needs less insulin infusion. Thus it is useful if we consider insulin sensitivity as input of fuzzy controller.

In our system (patient), there are two inputs: meal intake and delivered insulin. It is important to mention that only delivered insulin is controlled by output of designed fuzzy controller that can be implemented in insulin pump. This issue is caused the controller has not proctor on blood

glucose when the patient suffers a severe drop in blood glucose (hypoglycemic state). For this reason we suggest design of fuzzy controller with two outputs (insulin and glucose) and so an insulin pump in which there are two syringes and two reservoirs (one for insulin and one for glucose). With using this idea, the pump can mimic insulin secretion natural algorithm and can be as an artificial pancreas.

REFERENCES

- [1] B. M. Frier, Ph. E. Cryer "Hypoglycemia, In *International Textbook of Diabetes Mellitus*", 2nd ed., John Wiley and Sons Ltd, Chichester, England, 1997, pp. 1193-1214.
- [2] B. H. Ginsberg, "How accurate are accuracy measurements of continuous glucose sensing", *Diabetes Technology and Therapeutics*, vol. 7, 2005, pp. 673-675.
- [3] G. M. Steil, A. E. Panteleon, and K. Rebrin, "Closed-loop insulin delivery—the path to physiological glucose control," *Advanced Drug Delivery Reviews*, vol. 56, no. 2, 2004, pp. 125–144.
- [4] A. Wahab, Y. K. Kong, C. Quek, "Model reference adaptive control on glucose for the treatment of diabetes mellitus", 19th *IEEE Int. Symp. on computer-based medical systems (CBMS)*, 2006, pp. 315-320.
- [5] R. Hovorka *et al.*, "Nonlinear model predictive control of glucose concentration in subjects with type 1 diabetes", *Physiol. Meas.*, vol. 24, no. 4, 2004, pp. 905-920.
- [6] "The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus", *N. Engl. J. Med.*, vol. 329, 1993, pp. 977-986, DCCT.
- [7] L. Kardar, A. Fallah, SH. Gharibzadeh and F. Mozarzadeh, "Application of fuzzy logic controller for intensive insulin therapy in type 1 diabetic mellitus patients by subcutaneous route", *WSEAS Trans. on Syst. and Control*, Iss. 9, vol. 3, 2008, pp. 712-721.
- [8] "Standards of medical care for patients with diabetes mellitus", *Diabetes Care*, vol. 25, 2002, pp. 533-549. ADA.
- [9] R. N. Bergman, L. Philips, C. Cobelli, "Physiological evaluation of factors controlling glucose tolerance in man", *J. Clin. Invest.*, vol. 68, 1981, pp. 1456-1467.
- [10] M. E. Fisher, "A semiclosed-loop algorithm for control of blood glucose levels in diabetics", *IEEE Trans. Biomed. Eng.* vol. 38, no. 1, 1991, pp. 57-61.
- [11] Sh. Yassini, M. B. Naghibi-sistani and A. Karimpour, "Active insulin infusion using fuzzy-based closed-loop control", *Proc. 3rd Int. Conf. on Intell. Syst. and knowledge Eng.*, 2009, pp. 429-434.
- [12] G. Pacini and R. N. Bergman, "MINMOD: a computer program to calculate insulin sensitivity and pancreatic responsivity from the frequently sampled intravenous glucose tolerance test". *Comput. methods and programs in Biomed.*, vol. 23, 1986, pp. 113–122.
- [13] L. Dejournett, "Essential elements of the native glucoregulatory system, which, if appreciated, may help improve the function of glucose controllers in the intensive care unit setting", *J. diabetes science and technology*, Iss. 1, vol. 4, 2010, pp. 190-198.